EtherWM: Motion-Guided Flow Correction for Long-Horizon Robot Video Prediction

Anonymous CVPR submission

Paper ID 0000

Abstract

The visual consequences of robot motion depend on the current scene state and preceding interactions. We target minute-scale autoregressive video prediction conditioned on recorded execution trajectories. EtherWM explicitly models action-conditioned state transitions and uses their predictions to guide video generation. Its asymmetric two-tower architecture pairs a lightweight Motion Tower with a LoRA-adapted MiniMax-H3 generator. Both operate in the frozen H3-VisualVAE's latent space. Temporal latent differences supervise the Motion Tower, while flow targets supervise the generator's adapters and conditioning modules. At each denoising step, the disagreement between the forecast increments and H3's implied transitions yields an analytic velocity correction with a learned strength. A bounded short-term reference bank (H1) and long-term archive (H2) carry interaction context across chunks. Training on generated prefixes exposes both towers to the evolving context of their own rollouts. EtherWM ranks first on five of seven WorldArena metrics among the compared models, and qualitative comparisons extend to 110 seconds.

1. Introduction

The same upward gripper motion can lift a cup or leave it on the table, depending on whether a grasp has been established. During transport and release, the object's motion continues to depend on its interaction state. In long-horizon video prediction, each predicted outcome becomes the context for subsequent interactions, allowing an incorrect transition to affect many later chunks. We target continuous minute-scale autoregressive prediction from an initial visual context and recorded execution trajectories, without subsequent real-video resets. Consistency requires trajectory following, persistent object identity, and coherent interactions across grasping, transport, and release.

Robot world models combine trajectory conditioning with memory to generate extended interactions [3, 11, 14]. Temporal latent increments provide an explicit representation of visual change: DeltaWorld learns action-conditioned increments [4], and LaMo uses motion priors to guide video diffusion [8]. We ask how a transition forecast can guide denoising throughout a rollout while retaining interaction history. Figure 1 illustrates the resulting gains in trajectory following and object appearance.

We introduce **EtherWM**, an asymmetric two-tower framework for **explicit state-transition prediction and flow correction**. A lightweight Motion Tower forecasts latent increments from history, execution trajectories, and memory. MiniMax-H3 [12], adapted through LoRA [5], generates video. At each denoising step, the analytic gradient of their transition disagreement corrects H3's velocity with learned strength. The corrected flow objective jointly trains the Motion Tower, gate, and trainable H3 adapters. A short-term reference bank (H1), a bounded long-term archive (H2), and generated-prefix training [1, 7, 14] sustain this prediction–correction process across chunks.

Our contributions are threefold:

- **Explicit action-conditioned state transitions.** An asymmetric two-tower framework learns latent state changes from visual history and execution trajectories, with distinct supervision for transition prediction and video generation.
- **Learnable gradient correction.** We derive an analytic velocity correction from transition disagreement and learn its strength jointly with the Motion Tower and H3 adapters through corrected flow matching.
- **Memory-coupled long-horizon rollout.** Both towers read bounded H1/H2 memory and train on generated prefixes, carrying the same prediction and correction process across minute-scale rollouts.

2. Related Work

2.1. Robot World Models

Action-conditioned video models predict visual futures from robot trajectories. Ctrl-World combines frame-level action conditioning, multi-view prediction, and pose-

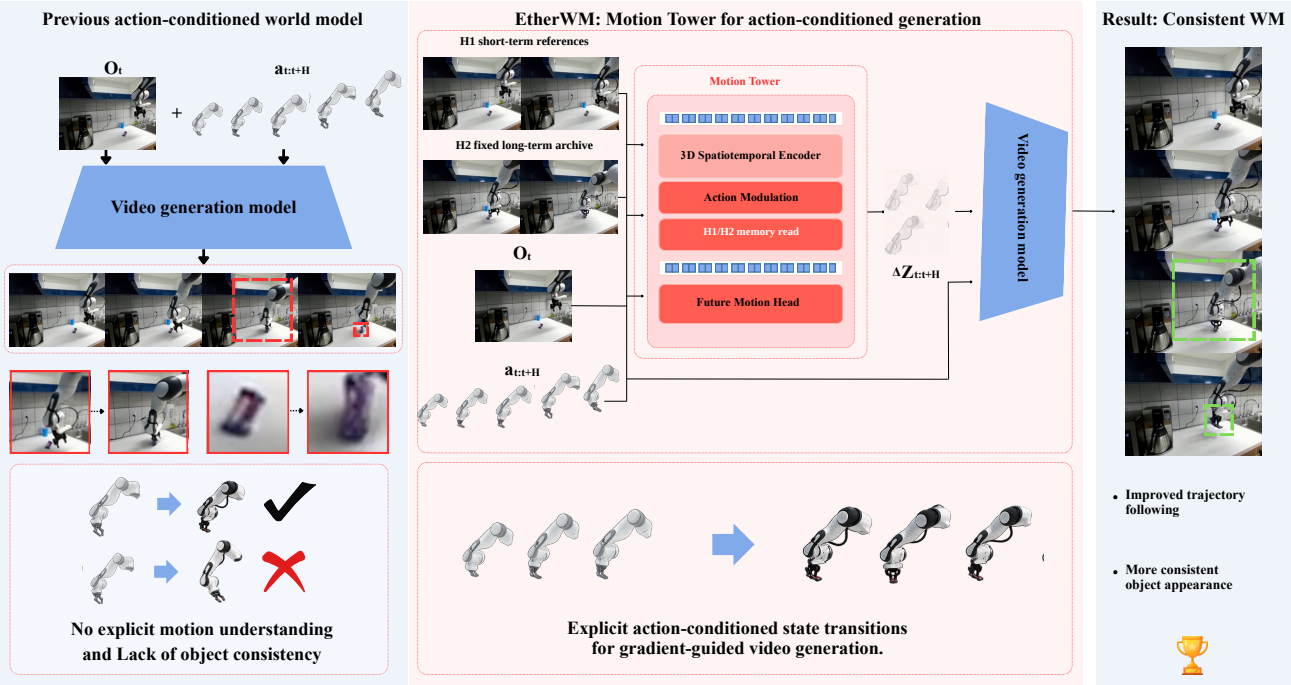

Figure 1. **Qualitative comparison of long-horizon action-conditioned robot video prediction.** Left: the previous world model departs from the recorded arm trajectory and exhibits object appearance drift. Center: EtherWM forecasts latent changes from recent visual history, H1/H2 memory, and execution trajectories, then guides H3 generation through gradient-based velocity correction. Right: EtherWM follows the robot trajectory more closely and better preserves object appearance in the illustrated rollout.

conditioned memory retrieval [3]. SolarWM adapts pre-trained video backbones while preserving their native representations and objectives [6]. Explicit transition models further structure the relationship between motion and scene evolution. DeltaWorld predicts action-conditioned latent increments with interaction-aware latent alignment [4]. Robot-Factored World Models expose controller-derived nominal trajectories, rendered robot geometry, and depth cues to the video generator [9].

EtherWM adapts MiniMax-H3 [12] to recorded execution trajectories through LoRA [5] and trainable conditioning modules. Its Motion Tower supplies a transition forecast before sampling begins; a learned-strength gradient correction connects that forecast to H3 at every denoising step. This coupling makes transition prediction a component of the video-generation process.

2.2. Motion Priors and Generative Guidance

Flow matching learns a vector field between noise and data [10]; its clean-data estimate provides an interface for latent-space constraints. LaMo learns a spatial motion predictor from ground-truth latent differences [8]. Backbone training supervises spatially averaged motion drift. During sampling, the frozen predictor reads the current clean-video estimate and supplies a motion-consistency gradient with prescribed guidance strength.

EtherWM forecasts changes once per chunk from the causal visual prefix, recorded execution trajectory, and H1/H2 memory. Holding this forecast fixed gives an analytic gradient of its disagreement with H3's implied temporal differences. The correction operates during training and sampling: the corrected flow objective learns its strength and optimizes the Motion Tower, memory readers, and H3's trainable adapters. This connects history-derived transition prediction to the video-generation vector field.

2.3. Memory and Long-Horizon Rollout

Long-horizon prediction couples historical context with training on accumulated model outputs. WALL-SS combines recent and compressed distant memory with scale-wise dream forcing and on-policy alignment [14]. RMWAM learns context-dependent utility for recent-state and event memories through counterfactual access [11]. MemWorld also studies memory-augmented, action-conditioned world modeling [15]. Self Forcing trains on autoregressively generated context using video-level supervision and gradient truncation [7]. Persistent Robot World Models optimizes robot-video rollouts through reward-contrasted de-

noising of candidate continuations from a common state [1].

EtherWM shares dense recent history and the H1/H2 banks between transition prediction and video generation. Completed chunks update this context, while generated-prefix training exposes the Motion Tower, correction gate, and H3 adapters to accumulated predictions. Historical evidence determines the transition forecast, the forecast guides sampling, and each sampled chunk supplies context for the next prediction.

3. Method

EtherWM couples explicit state-transition prediction to video generation through an analytic gradient correction with learned strength (Fig. 2). A Motion Tower forecasts latent increments, LoRA-adapted MiniMax-H3 generates the next video chunk, and shared H1/H2 memory carries interaction context across the rollout.

3.1. Trajectory-Conditioned Video Prediction

Given an initial image and recorded execution trajectories, we target minute-scale autoregressive prediction with MiniMax-H3 [12]. Its frozen H3-VisualVAE uses 24 channels with spatial compression by 16 and native temporal compression by 4. We preserve its chunk boundaries and the transformer's $1 \times 2 \times 2$ patchification. Standard chunks contain $K = 5$ latents aligned with 17 execution states; tails contain $K = 2$ latents and five valid states. The native temporal groups are $[1, 4, 4, 4, 4]$, with the first two used for tails.

For chunk j, H_j contains at most five recent chunks (25 latents); A_j denotes the aligned execution trajectory, and $\mathcal{M}^S_j, \mathcal{M}^L_j$ denote the H1 and H2 banks (Sec. 3.4). With h_j the last latent in H_j, the target is

$$Z_j = (z_{j,1}, \ldots, z_{j,K}), \qquad z_{j,k} \in \mathbb{R}^{24 \times H_z \times W_z}. \tag{1}$$

Here (H_z, W_z) is the VAE spatial grid at the chosen video resolution. H3 retains initial-image, text, and view conditions. We omit the chunk index below.

3.2. Explicit Action-Conditioned State Transitions

The Motion Tower (MT) predicts K latent increment fields once per chunk from recent history H, the recorded trajectory A, and both memory banks $\mathcal{M}^S$ and $\mathcal{M}^L$. Its *History Encoder* pools dense history to an 8×10 grid; spatiotemporal convolutions summarize the final state, recent change, and temporal mean. Independent projections encode ordered native action groups. *Action modulation* forms future queries Q, which read H1/H2 tokens through memory attention. A *Difference Head* upsamples and refines each forecast using the final history state and recent difference. The forecast remains fixed while H3 denoises the chunk.

We supervise the forecast $\widehat{\Delta} = \mathrm{MT}_\phi(H, A, \mathcal{M}^S, \mathcal{M}^L)$ through anchored temporal differences:

$$\begin{aligned} \mathcal{D}_h(Y) &= (y_1 - h,\ y_2 - y_1, \ldots, y_K - y_{K-1}), \\ \Delta^* &= \mathcal{D}_h(Z^*). \end{aligned} \tag{2}$$

We average Smooth-L_1 over valid latent elements:

$$\mathcal{L}_\Delta = \mathrm{mean}_{\mathrm{valid}} \left[\rho\left(\widehat{\Delta} - \Delta^* \right) \right]. \tag{3}$$

The predicted state is $\widehat{z}_k = h + \sum_{i=1}^{k} \widehat{\Delta}_i$. This objective directly supervises the transition forecast; the separate flow objective trains H3 to realize video consistent with that forecast.

3.3. Learnable Gradient Correction

H3 uses the dataward flow convention [10, 12]:

$$x_t = tZ^* + (1-t)\epsilon, \qquad v^* = Z^* - \epsilon. \tag{4}$$

H3 uses packed action tokens and temporally aligned adaptive layer normalization. Writing θ_0 for frozen backbone weights and η for trainable adapters, its base velocity is $v_{\mathrm{base}} = f_{\theta_0,\eta}(x_t, t, H, A, \mathcal{M}^S, \mathcal{M}^L)$. With $s = 1 - t$, the clean estimate is

$$W = x_t + s v_{\mathrm{base}}. \tag{5}$$

Transition disagreement defines

$$r = \mathcal{D}_h(W) - \widehat{\Delta}, \qquad E = \tfrac{1}{2}\|r\|_2^2, \tag{6}$$

$$g = \nabla_W E = (r_1 - r_2, \ldots, r_{K-1} - r_K, r_K). \tag{7}$$

Since $\nabla_{v_{\mathrm{base}}} E = sg$, the *Velocity Correction* block produces $\Delta v = -s\alpha_\psi \odot g$:

$$\boxed{v_{\mathrm{corr}} = v_{\mathrm{base}} + \Delta v.} \tag{8}$$

This yields $W_{\mathrm{corr}} = W - s^2 \alpha_\psi \odot g$. A small 3D convolutional gate predicts $\alpha_\psi = \alpha_{\max} \tanh G_\psi(U)$, where U contains channelwise residual RMS, the mean absolute transition magnitudes of H3 and MT, and s. The gain has shape $1 \times K \times H_z \times W_z$ per sample and broadcasts across channels. Its final layer is zero-initialized.

Correction acts on the unpatchified target-video velocity. The native sampler sets $t_i = 1 - \sigma_i$ and $x_{i+1} = x_i + (\sigma_i - \sigma_{i+1}) v_{\mathrm{corr}}$ on a decreasing noise grid. The analytic temporal-difference gradient requires no additional H3 forward pass or inference backpropagation.

3.4. Dual-Horizon Memory and Autoregressive Rollout

Both towers read a short-term reference bank $\mathcal{M}^S$ (H1) and a fixed-capacity long-term archive $\mathcal{M}^L$ (H2).

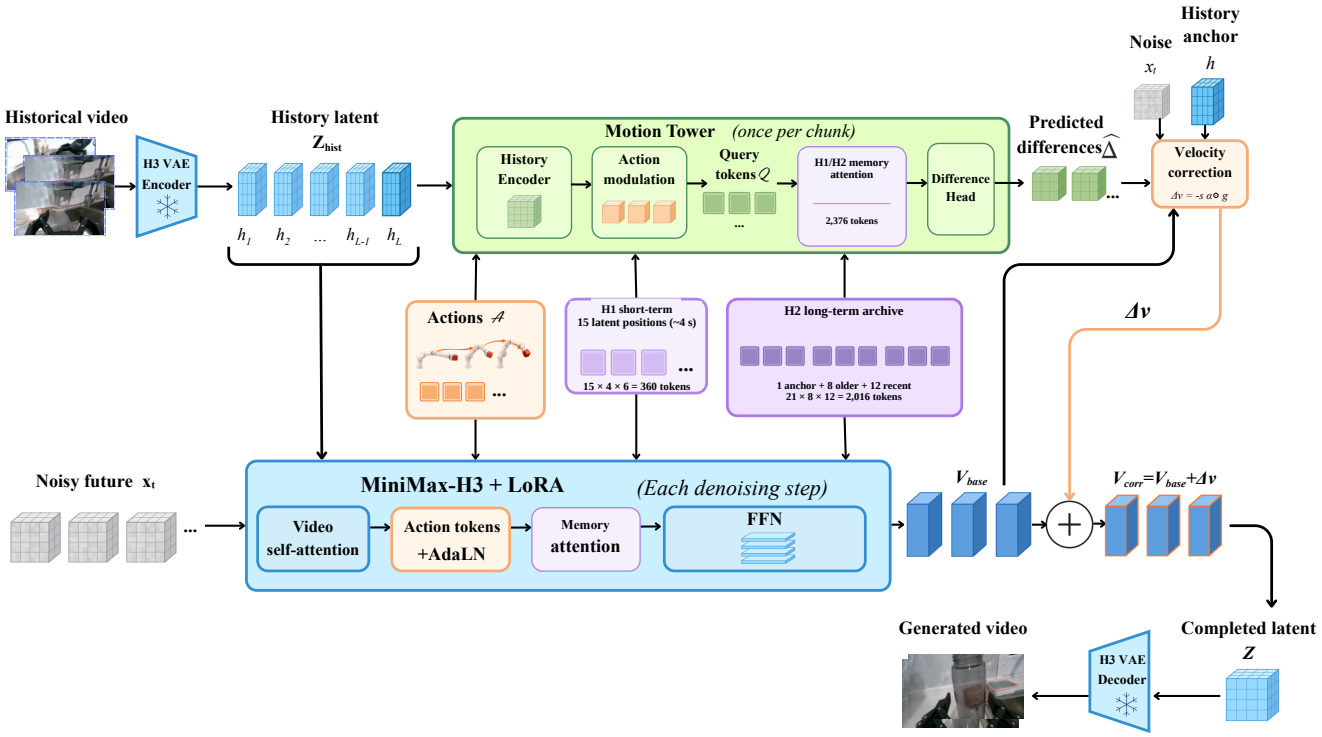

Figure 2. **EtherWM overview.** The Motion Tower predicts latent state increments once per chunk from recent visual history, execution trajectories, and the H1/H2 banks. At each denoising step, disagreement with H3's implied transitions yields an analytic velocity correction with learned strength. Both towers read H1 short-term references and the bounded H2 archive. Each completed chunk updates the shared history and memory for the next forecast.

Short-term references (H1). We pool each of the latest 15 H3 latent positions to a 4×6 grid, yielding 360 tokens of width 256 with spatial and relative-time embeddings. With 5 FPS input and a temporal stride of four, these positions span approximately 11 seconds of recent interactions.

Long-term archive (H2). H2 retains 21 temporal slots: one initial observed anchor, eight stratified earlier positions, and twelve recent positions. Each slot contains an 8×12 grid, for 2,016 records total. Each record combines 24 H3 latent channels with relative time, two spatial coordinates, and observation origin, then projects to width 256. Unfilled slots are masked. H2 associates normalized visual features by mutual nearest-neighbor matching. A gated write accepts matches above a similarity threshold τ and a nearest-neighbor margin μ.

Memory reads and updates. MT queries both banks once per chunk. H3 reads them through four-head memory attention at every fifth transformer block, with a zero-initialized output projection and a bounded residual gate. The banks remain fixed during denoising. The completed latent $\widehat{Z}_j$ updates recent history and both banks:

$$(H_{j+1}, \mathcal{M}^S_{j+1}, \mathcal{M}^L_{j+1}) = \mathcal{U}(H_j, \mathcal{M}^S_j, \mathcal{M}^L_j, \widehat{Z}_j). \quad (9)$$

Generated-prefix training fills the banks with sampled chunks, as at inference, and detaches stored content at chunk boundaries. The first chunk uses native H3 generation from the initial image; transition correction starts once a completed history chunk is available.

3.5. Joint Optimization

We train MT, the correction gate, and H3 LoRA and action/view adapters while keeping the pretrained backbone, H3-VisualVAE, and text encoder frozen. H1/H2 use fixed slot capacities and update schedules; their lightweight association, write, and read projections learn through the transition and flow objectives.

After warming up MT with $\mathcal{L}_\Delta$, we enable correction and jointly optimize

$$\mathcal{L} = (1-\beta)\mathcal{L}_{\text{base}} + \beta\mathcal{L}_{\text{corr}} + \lambda_\Delta \mathcal{L}_\Delta + \lambda_r \mathcal{L}_{\text{reg}}. \quad (10)$$

Here $\mathcal{L}_{\text{base}}$ and $\mathcal{L}_{\text{corr}}$ compare the respective velocities with v^* using the same valid-element mask and mean normalization; $\mathcal{L}_{\text{reg}} = \text{MSE}(v_{\text{corr}}, v_{\text{base}})$. Corrected-flow gradients

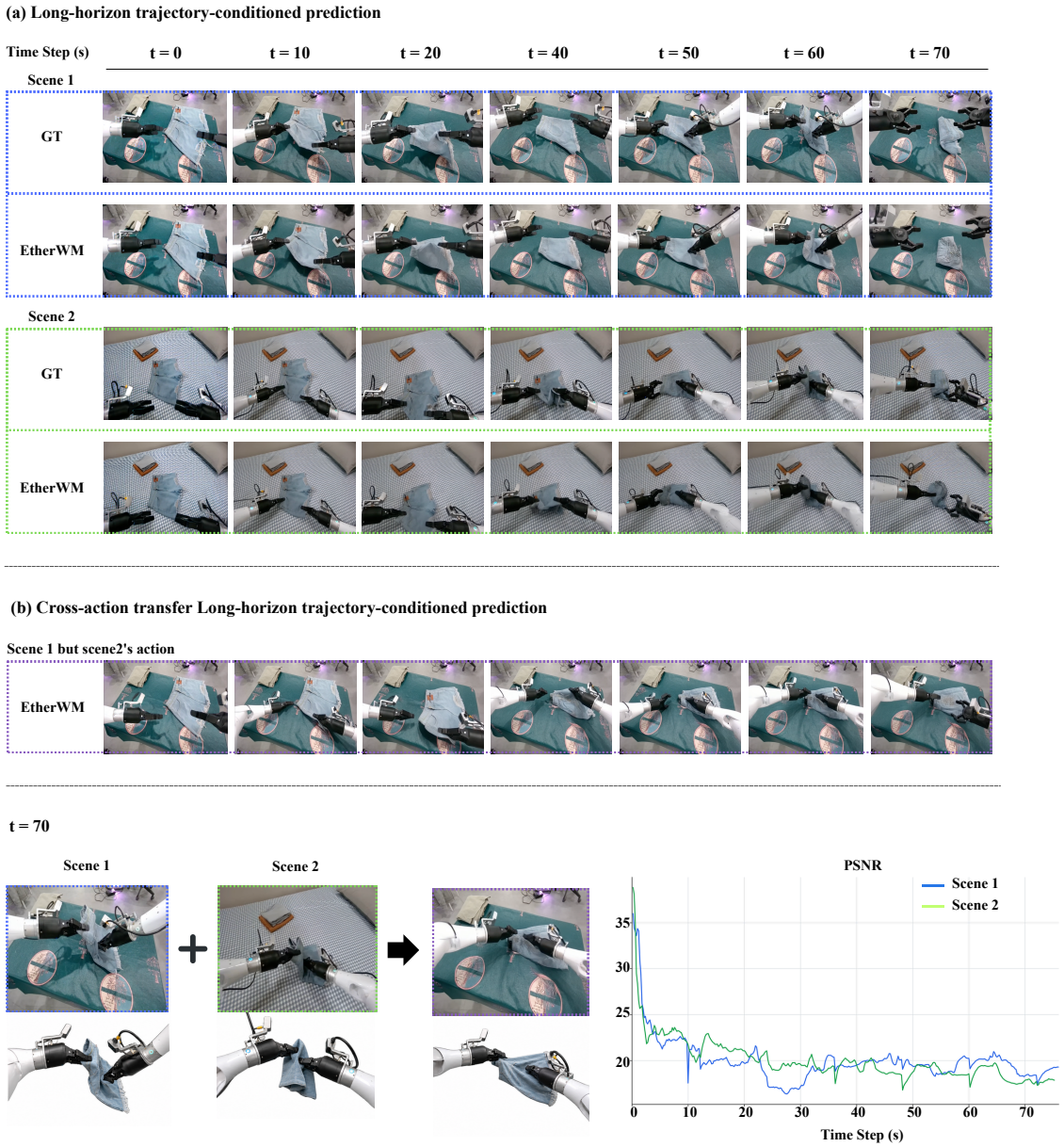

Figure 3. **Representative examples of long-horizon action following and cross-action transfer.** (a) Matched GT and EtherWM frames in two scenes over 70 seconds. (b) Scene 1 conditioned on Scene 2's execution trajectory. Enlarged views illustrate the transferred response at 70 seconds; PSNR curves measure fidelity for the two original scene–trajectory pairs.

Table 1. **WorldArena evaluation on RoboTwin 2.0** [2, 13]. Higher is better for all seven metrics. Bold and underlined values indicate the best and second-best results.

Model	Trajectory Accuracy ↑	Action Response Sensitivity ↑	Instruction Following ↑	Interaction Quality ↑	Photometric Consistency ↑	Depth Accuracy ↑	Semantic Alignment ↑
Genie Envisioner	6.79	1.09	20.28	20.52	20.06	86.63	85.44
Cosmos-Predict 2.5 (action)	29.45	1.33	58.40	55.00	<u>35.28</u>	88.62	88.79
IRASim	36.39	<u>5.26</u>	66.04	56.56	35.22	93.12	88.49
Ctrl-World	47.66	2.10	72.72	62.12	17.29	93.00	89.12
BWM	<u>64.36</u>	3.42	<u>82.48</u>	<u>73.76</u>	28.71	**97.56**	**90.59**
EtherWM	**67.38**	**12.35**	**84.62**	**76.83**	**57.21**	<u>97.19</u>	<u>89.54</u>

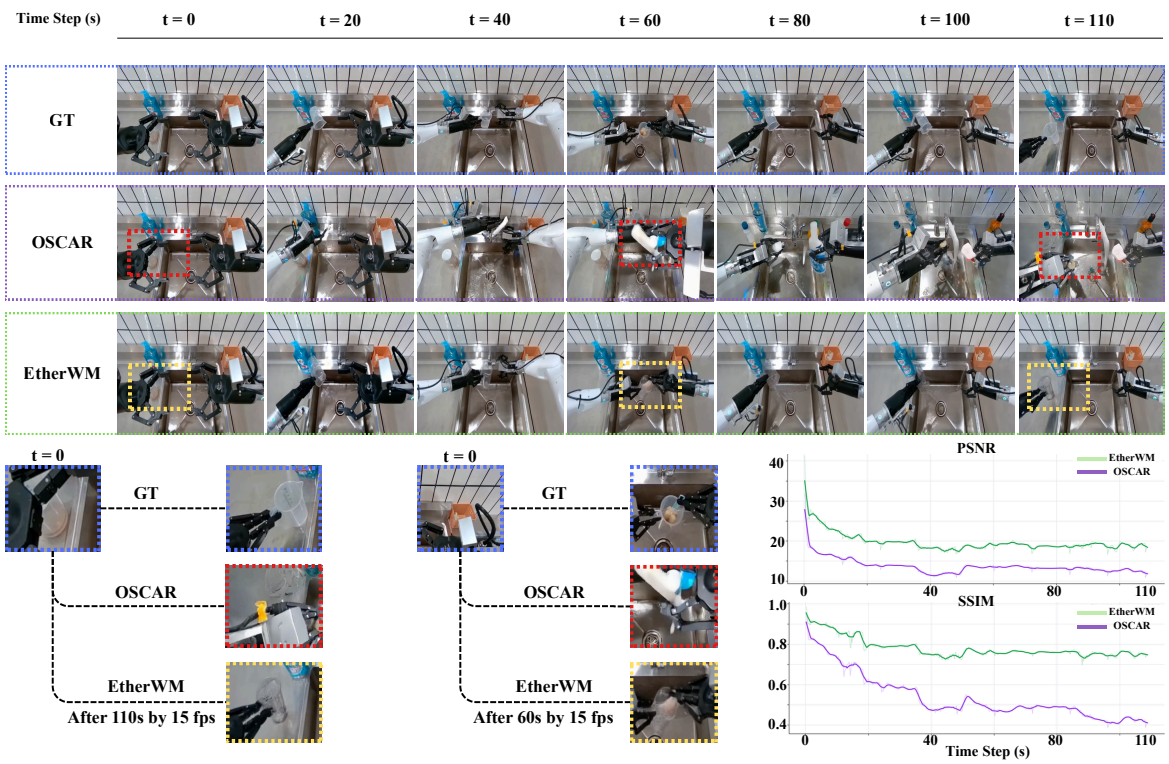

Figure 4. **110-second comparison with OSCAR-2B.** Matched GT, OSCAR-2B, and EtherWM frames show long-horizon trajectory following and object consistency. Enlarged regions highlight interactions at 60 and 110 seconds. PSNR and SSIM curves measure fidelity throughout the rollout (higher is better).

Table 2. **Quantitative component ablations.** Results on 500 held-out DROID and AgiBot episodes. ↑/↓ indicate higher/lower is better; best results in each column are bold.

Model	PSNR↑	Before 30s PSNR↑	After 30s PSNR↑	SSIM↑	LPIPS↓	Trajectory Accuracy↑
H3 + Action	18.21	20.56	15.37	0.52	0.31	45.02
EtherWM w/o Memory	19.85	21.79	17.95	0.69	0.16	**74.89**
EtherWM w/o Motion Tower	20.51	22.33	18.76	0.73	0.19	50.21
EtherWM	**21.99**	**22.71**	**21.26**	**0.79**	**0.13**	73.35

update MT, the gate, and H3 adapters. Both flow losses reuse one generator forward pass. Training and sampling use the scheduler's effective time, with correction arithmetic in FP32.

4. Experiments

Datasets and task coverage. We evaluate minute-scale prediction and action transfer on DROID 1.0.1 and AgiBot World Beta. The test set covers manipulation episodes from multiple scenes and includes both single-arm and dual-arm settings. Given an initial visual context, the model autoregressively predicts video conditioned on the recorded robot execution trajectory.

Backbone. EtherWM uses MiniMax-H3 as its pretrained video-generation backbone [12], adapted through LoRA [5] and trainable action/view conditioning.

Video and action representation. We resize videos to 512×768 (height $\times$ width) and downsample the original 30-FPS recordings to 5 FPS. Each prediction uses five history frames as context and predicts the next three frames. Each arm action is represented by the end-effector position (x, y, z), a 6D rotation representation, and gripper state, for 10 features per arm slot; the format supports up to two arm slots. Separate body-control channels represent the chassis, head, and waist. Actions and video frames are strictly aligned on the downsampled 5-FPS timeline.

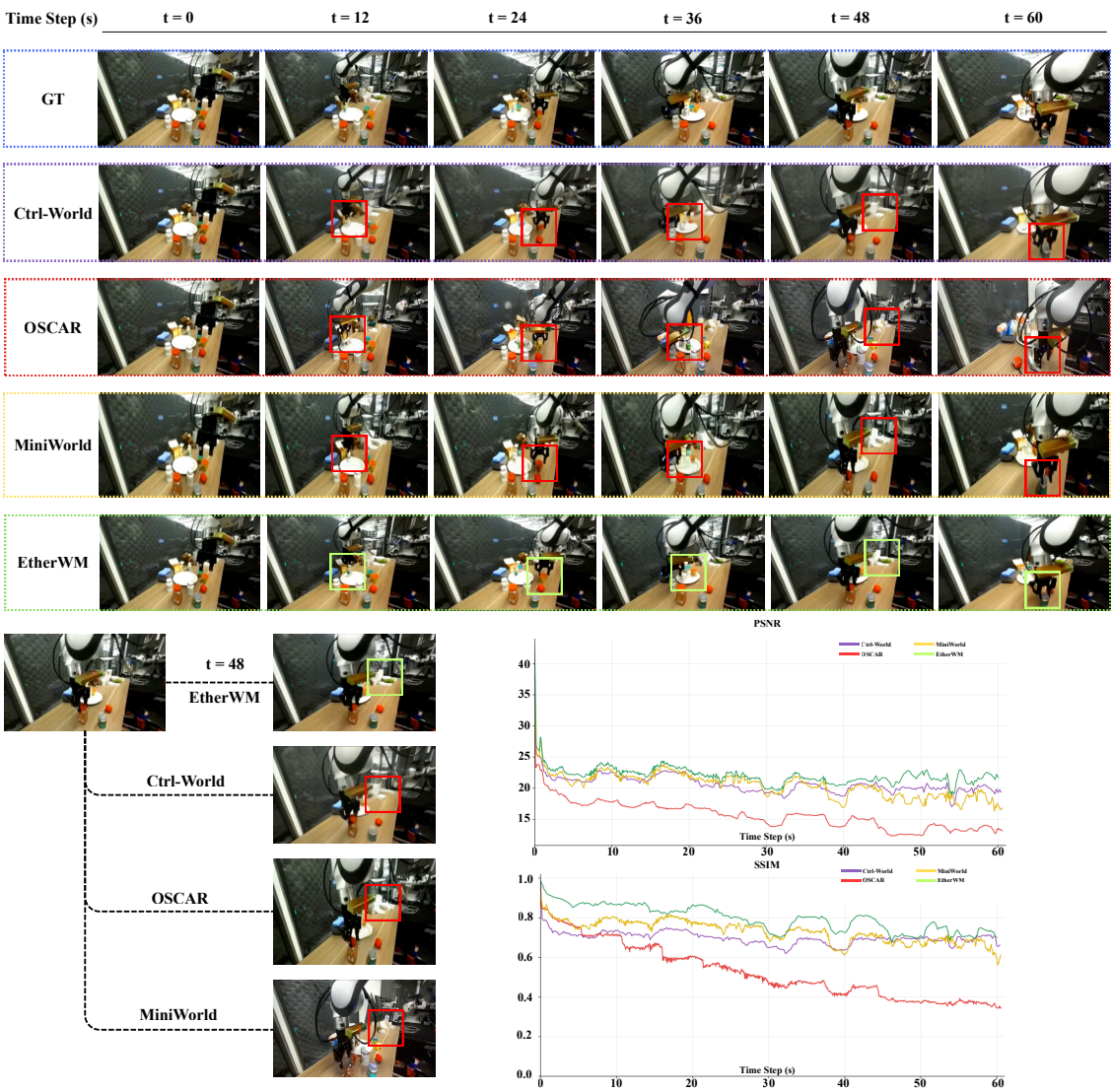

Figure 5. **Long-horizon qualitative and quantitative comparison on DROID.** Matched frames and fidelity curves compare Ctrl-World, OSCAR, MiniWorld, and EtherWM against GT. Insets show the gripper–bottle interaction at 12 seconds and background details at 48 seconds.

Training details. We pretrain for one epoch on the complete mixture of AgiBot World and DROID 1.0.1, followed by one epoch of fine-tuning on long-horizon data. We use a learning rate of 3×10^{-7} and gradient accumulation of 7.

Inference and evaluation. We use the native 32-step Euler sampler with noise-schedule shift 12 and guidance scale 1.0. For WorldArena, we evaluate the RoboTwin 2.0-adapted checkpoint on the complete official Clean-50 test set (50 tasks, 10 episodes per task). RoboTwin 2.0 is used for the WorldArena evaluation; long-horizon and component-ablation results are reported on DROID and AgiBot.

Data split. DROID and AgiBot are each divided into train/validation/test splits in a 7:2:1 ratio. The validation split is used for model selection, and the test split is reserved for final evaluation. The 8,654 minute-scale evaluation episodes are a subset of this held-out test split. WorldArena follows the official RoboTwin 2.0 split.

Trajectory Accuracy. Following WorldArena [13], we extract image-normalized 2D gripper trajectories from generated and reference videos using SAM 3 bounding-box centers, with missing detections interpolated. The official evaluator selects the valid reference track with the largest spatial extent and computes the mean Euclidean distance

CVPR
#0000

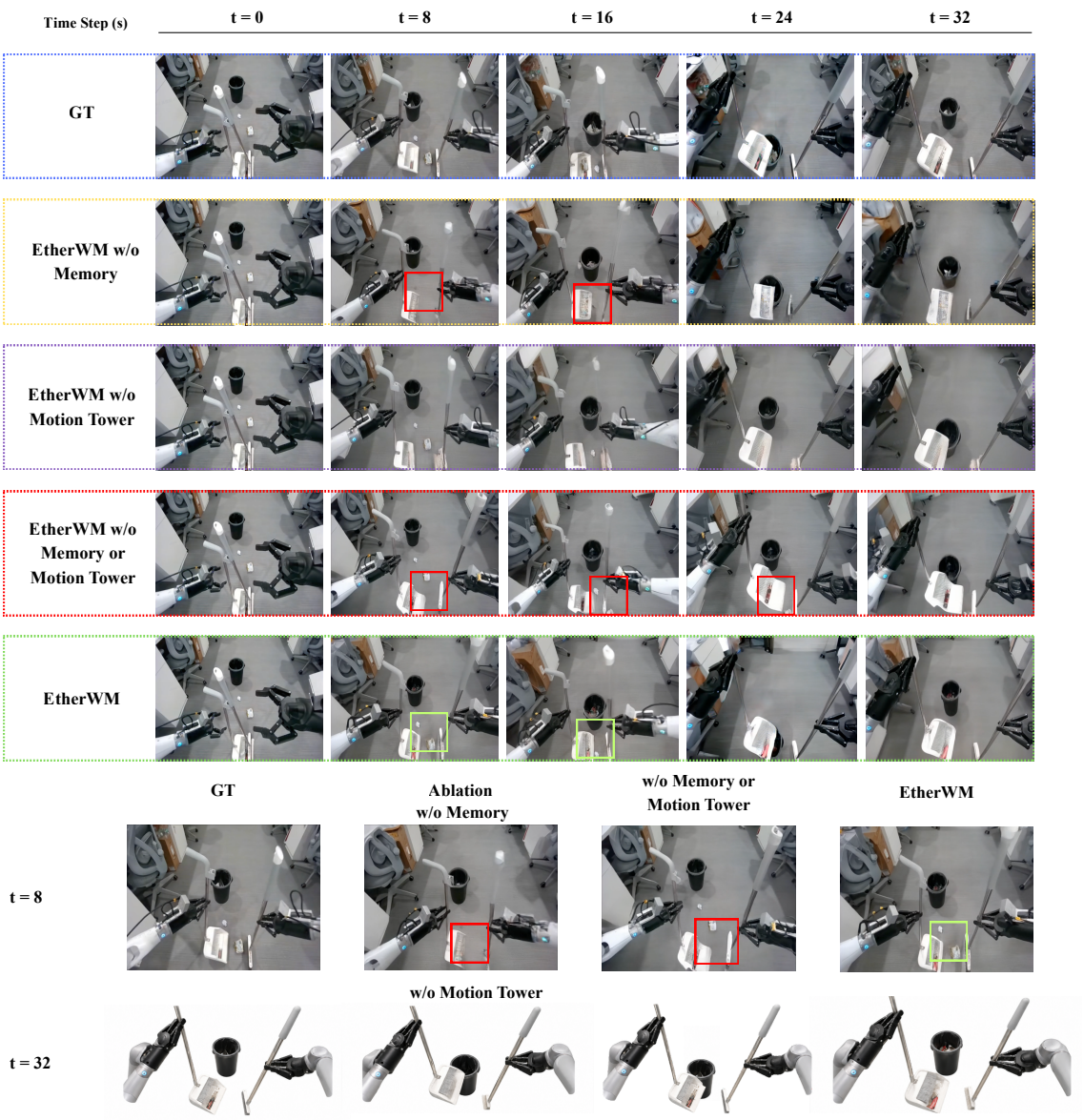

Figure 6. **Component ablations on an AgiBot manipulation task.** Top: GT and three model variants at 8-second intervals over a 32-second rollout. Bottom: enlarged comparisons of scene state at 8 seconds and arm–tool configuration at 32 seconds. Boxes highlight floor debris lost without memory and arm motion that deviates from GT without the Motion Tower.

d along the FastDTW alignment path. Its inverse, $1/d$, is normalized using WorldArena's fixed empirical bounds $[0, 40.8540]$ and clipped to $[0, 1]$. We report the mean normalized score multiplied by 100; higher values indicate better trajectory alignment. Predicted tracks with more than 90% missing detections receive zero.

4.1. Long-Horizon Action Following

Figure 3(a) shows trajectory following and object consistency through 70 seconds. Across 8,654 held-out minute-scale episodes, EtherWM achieves PSNR 21.98, SSIM 0.802, LPIPS 0.096, FVD 98.0435, and Trajectory Accuracy 63.51 under matched scene–trajectory pairing. These episodes account for approximately 10% of the test split. To evaluate cross-episode action-follow generalization, we randomly pair episode indices: the initial visual context comes from a receiving episode, while the action sequence comes from a different source episode. The generated rollout is evaluated against the gripper trajectory extracted from that source episode, whose action sequence is transferred to the receiving context. Over the same 8,654 held-out episodes, EtherWM achieves a Trajectory Accuracy of

Table 3. Trajectory Accuracy under matched and index-randomized cross-episode action pairing. Matched actions are evaluated against their own episode trajectory; for transferred actions, each receiving episode's initial view is paired with an action sequence from another episode, and the generated gripper track is evaluated against the source episode trajectory. Δ is the matched-action score minus the transferred-action score.

Model	Matched ↑	Transferred ↑	Δ ↓
MiniWorld	62.32	43.65	18.67
Ctrl-World	58.73	10.02	48.71
OSCAR	59.66	17.53	42.13
EtherWM	**63.51**	**61.02**	**2.49**

61.02 under transferred actions, compared with 63.51 under matched pairing. Figure 3(b) and Fig. 7 show qualitative examples of these cross-episode rollouts.

Table 3 compares trajectory accuracy across models under the same index-randomized episode pairing. EtherWM achieves 61.02 against the source-episode reference trajectory, exceeding all compared models and decreasing by only 2.49 points from its matched-action score of 63.51. The result demonstrates action-follow generalization to cross-episode scene–action pairings in the held-out test set.

Figure 7. **Qualitative action transfer across DROID episodes.** Each example pairs a receiving scene's initial frame with an execution trajectory transferred from another episode. Columns show frames sampled over the rollout; red boxes highlight interaction regions where the predicted arm motion and scene response can be inspected.

We evaluate cross-episode action-transfer rollouts with Qwen2.5-VL-72B (Table 4). From the DROID test set, we select 500 episodes spanning scenes from simple to complex. For each model and episode, we extract video frames and score them individually with the same fixed prompt, which instructs the evaluator to judge only the current frame; model identities are hidden. The prompt requests 0–10 scores for (i) scene and object consistency, including appearance, shape, and spatial relations; (ii) robot-structure consistency, including links, joints, and gripper topology; and (iii) interaction plausibility, including physical contact and support while penalizing penetration or unsupported floating. We manually review 50 episodes, corresponding to 200 videos across the four models. EtherWM scores 8.31, 9.32, and 7.31, compared with 7.75, 7.85, and 6.29 for Ctrl-World. These frame-wise ratings complement trajectory measures by assessing appearance, robot structure, and interaction quality; they do not measure temporal consistency.

Table 4. Frame-wise Qwen2.5-VL-72B assessment of cross-episode action-transfer rollouts on 500 DROID test episodes. Scores range from 0 to 10; higher is better.

Model	Scene/object ↑	Robot structure ↑	Interaction ↑
Ctrl-World	7.75	7.85	6.29
MiniWorld	7.46	7.85	6.08
OSCAR	6.60	7.08	5.29
EtherWM	**8.31**	**9.32**	**7.31**

4.2. Long-Horizon Comparison with OSCAR

We compare EtherWM with OSCAR-2B [16] on a 110-second manipulation sequence (Fig. 4). The cup is outside the arm camera's direct view, and the brush is intermittently visible, so the models must carry interaction state across gaps in visual evidence.

Figure 4 compares the ground truth (GT), OSCAR-2B, and EtherWM at matched points throughout the rollout. The qualitative frames and enlarged interaction regions show that EtherWM more closely follows the demonstrated arm motion and better preserves manipulated-object consistency over time. The accompanying PSNR and SSIM curves provide frame-level fidelity measurements: EtherWM remains above OSCAR-2B for most of the displayed horizon, with the advantage persisting into late rollout.

4.3. Long-Horizon DROID Comparison

Figure 5 compares EtherWM with Ctrl-World, OSCAR, and MiniWorld on a DROID episode. EtherWM most closely reproduces the gripper–bottle interaction at 12 seconds and is the only compared model that retains the green vegetables visible in the GT background at 48 seconds. PSNR and SSIM support the visual comparison over the displayed rollout.

4.4. WorldArena Quantitative Evaluation

EtherWM leads on five of seven metrics on RoboTwin 2.0 (Table 1), including Trajectory Accuracy (67.38), Action Response Sensitivity (12.35), and Photometric Consistency (57.21). Action Response Sensitivity is 2.35 times the best baseline score (IRASim, 5.26). EtherWM ranks second in Depth Accuracy and Semantic Alignment.

4.5. Component Ablations

We evaluate component contributions on 500 randomly selected short- and long-horizon test episodes, with DROID 1.0.1 and AgiBot World Beta contributing 250 episodes each (Table 2). The memory ablation removes H1, the short-term reference bank, and H2, the long-term archive, together as EtherWM's dual-horizon memory system.

Removing this memory system primarily affects long-horizon visual fidelity. Compared with full EtherWM, EtherWM w/o Memory lowers PSNR from 21.99 to 19.85 and SSIM from 0.79 to 0.69, while LPIPS increases from 0.13 to 0.16. After 30 seconds, its PSNR is 17.95, 3.31 dB below EtherWM (21.26). In Figure 6, floor debris progressively disappears without memory by 8 seconds, whereas it remains visible in the full model. This result reflects the role of H1 and H2 in retaining recent interaction details and earlier scene context across chunks.

The trajectory results show a different pattern: EtherWM w/o Memory scores 74.89, 1.54 points above the full model's 73.35. Thus, this ablation supports the memory system's contribution to visual fidelity and scene consistency, while it does not show a gain in Trajectory Accuracy. By comparison, removing the Motion Tower reduces Trajectory Accuracy to 50.21; Figure 6 also shows its arm–tool configuration deviating from GT at 32 seconds. These results associate the Motion Tower more directly with action following and the dual-horizon memory system with preserving visual context over long rollouts.

5. Conclusion

EtherWM makes the action–effect relation explicit in action-conditioned robot video prediction. A Motion Tower forecasts latent transitions from visual history, recorded execution, and retrieved memory; an analytic gradient correction uses these forecasts to guide MiniMax-H3's denoising field. H1 and H2 provide bounded short- and long-horizon references on H3's latent grid, and completed chunks update the archive during generated-prefix rollout. This design couples transition prediction and video generation across minute-scale autoregressive context.

References

[1] Jai Bardhan, Patrik Drozdik, Josef Sivic, and Vladimir Petrik. Persistent robot world models: Stabilizing multi-step rollouts via reinforcement learning, 2026. 1, 3

[2] Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, Weiliang Deng, Yubin Guo, Tian Nian, Xuanbing Xie, Qiangyu Chen, Kailun Su, Tianling Xu, Guodong Liu, Mengkang Hu, Huan-ang Gao, Kaixuan Wang, Zhixuan Liang, Yusen Qin, Xiaokang Yang, Ping Luo, and Yao Mu. RoboTwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation. *arXiv preprint arXiv:2506.18088*, 2025. 5

[3] Yanjiang Guo, Lucy Xiaoyang Shi, Jianyu Chen, and Chelsea Finn. Ctrl-World: A Controllable Generative World Model for Robot Manipulation, 2025. 1, 2

[4] Boyuan Hou, Xiaoge Cao, Chaofan Zhang, Shuo Wang, and Shaowei Cui. DeltaWorld: Physically consistent interactive world simulators via action-conditioned latent increment learning, 2026. 1, 2

[5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. 1, 2, 6

[6] Junchao Huang, Guian Fang, Shengju Qian, Xianghao Kong, Zhuoran Zhao, Wei Huang, Yihua Du, Zixin Zhang, Justin Cui, Yuchao Gu, Yukang Chen, Xinting Hu, Tianyu He, Shaoshuai Shi, Zhuotao Tian, Xin Wang, Mike Zheng Shou, and Li Jiang. SolarWM: Open Data and Scalable Training for Long-Horizon Video World Models, 2026. 2

[7] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion, 2025. 1, 2

[8] Bo Jiang, Depu Meng, Yihan Hu, Yichen Xie, Tianshuo Xu, and Wei Zhan. LaMo: Self-supervised latent motion priors for physical realism in video generation, 2026. 1, 2

[9] Byungjun Kim, Taeksoo Kim, Hyunsoo Cha, and Hanbyul Joo. Robot-factored world models via robot rendering, 2026. 2

[10] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023. 2, 3

[11] Cui Miao, Junhao Yu, Ping Liu, Ning Yang, Xuhua Chen, He Zheng, Xiaodong Wang, and Tao Zhang. RM-WAM: Reliable Memory-Augmented World Action Models for Long-Horizon Robot Manipulation, 2026. 1, 2

[12] MiniMax. MiniMax-H3: Official model and code release. Official model repository, 2026. 1, 2, 3, 6

[13] Yu Shang, Zhuohang Li, Yiding Ma, Weikang Su, Xin Jin, Ziyou Wang, Lei Jin, Xin Zhang, Yinzhou Tang, Haisheng Su, Chen Gao, Wei Wu, Xihui Liu, Dhruv Shah, Zhaoxiang Zhang, Zhibo Chen, Jun Zhu, Yonghong Tian, Tat-Seng Chua, Wenwu Zhu, and Yong Li. WorldArena: A unified benchmark for evaluating perception and functional utility of embodied world models. *arXiv preprint arXiv:2602.08971*, 2026. 5, 7

[14] Maeve Zhang, Rain Sun, Xiang Wang, Cyril Zhang, Shalfun Li, Meng Cao, Howard Lu, Ethan Chen, Harry Jhou, KZ Zheng, Lights Shi, Regis Cheng, Lorenzin, Robert Wang,

Victor Yao, Gody Li, Elise Mon, Yohann Tang, Ryan Yu, PS Zhang, Vincent Chen, Hang Su, Roy Gan, Hao Wang, and Qian Wang. WALL-SS: Scaling Long-horizon World Models via Next-Scale Autoregression, 2026. 1, 2

[15] Zirui Zheng, Jiaqian Yu, Xiongfeng Peng, jun shi, Mingyi Li, Chao Zhang, Weiming Li, Dong Wang, Huchuan Lu, and Xu Jia. Mem-World: Memory-Augmented Action-Conditioned World Models for Persistent Robot Manipulation, 2026. 2

[16] zywu2115. OSCAR-2B. Hugging Face model card, n.d. 9